

## Chapter Five

### Higher Order Linear Differential Equations

#### Section-5.1

##### Wronskian

##### 5.1.1 Wronskian

Let us assume that each of the functions  $f_1, f_2, \dots, f_n$  is differentiable at least  $(n-1)$  times in the interval  $a \leq x \leq b$ . Then the determinant

$$W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix}$$

in which primes denotes the derivatives, is called wronskian of these  $n$  functions.

##### 5.1.2 Linear Dependence and Independence

**Linear dependence:** Given the  $n$  functions  $f_1, f_2, \dots, f_n$ , are said to be **linearly dependent** in some interval  $a \leq x \leq b$ , if there exists constants  $c_1, c_2, \dots, c_n$ , not all zero, such that

$$c_1 f_1 + c_2 f_2 + \dots + c_n f_n = 0 \quad \dots (1)$$

**Example:** Is the set of functions  $f_1(x)$ ,  $f_2(x)$ ,  $f_3(x)$ ,  $f_4(x)$ ,  $f_5(x)$  are linearly dependent on the interval  $(0, \infty)$ , where  $f_1(x) = x^2 + x - 1$ ,  $f_2(x) = 5\sqrt{x} - 2$ ,  $f_3(x) = 5\sqrt{x} - 2x$ ,  $f_4(x) = 2x^3$ ,  $f_5(x) = x^2$ ?

The functions  $f_1(x)$ ,  $f_2(x)$ ,  $f_3(x)$ ,  $f_4(x)$ ,  $f_5(x)$  are linearly dependent on the interval  $(0, \infty)$ , because  $f_3$  can be written as a linear combination of the other functions  $f_1$ ,  $f_2$ ,  $f_4$  and  $f_5$ .

i.e.,  $f_3(x) = -2 \cdot f_1(x) + 1 \cdot f_2(x) + 0 \cdot f_4(x) + 2 \cdot f_5(x)$

**Linear independence:** Given the  $n$  functions  $f_1, f_2, \dots, f_n$ , are said to be **linearly independent** in some interval  $a \leq x \leq b$ , if they are not linearly dependent there. That is, the  $n$  functions  $f_1, f_2, \dots, f_n$ , are said to be **linearly independent** in some interval  $a \leq x \leq b$ , if there exists constants  $c_1, c_2, \dots, c_n$ , such that

$$c_1 f_1 + c_2 f_2 + \dots + c_n f_n = 0 \quad \dots (1)$$

implies that  $c_1 = c_2 = \dots = c_n = 0$ .

**Example:** Is the set of functions  $f_1(x)$ ,  $f_2(x)$  are linearly independent on the interval  $(-\infty, \infty)$ , where  $f_1(x) = x^2 + x$ ,  $f_2(x) = x^3$ ?

The functions  $f_1(x)$ ,  $f_2(x)$  are linearly independent on the interval  $(-\infty, \infty)$ , because  $f_1$  and  $f_2$  cannot be written as linear combination. That is,  $c_1 f_1(x) + c_2 f_2(x) = 0$  is only possible when  $c_1 = c_2 = 0$ .

##### 5.1.3 Linear Dependence and Independence Theorems

**Theorem-1:** Let  $y_1, y_2, \dots, y_n$  be  $n$  solutions of the homogeneous linear  $n$  th-order differential equation on an interval  $I$ . Then the set of solutions is linearly independent on  $I$  if and only if  $W(y_1, y_2, \dots, y_n) \neq 0$  for all  $x$  in the interval.

**Theorem-2:** Let  $y_1, y_2, \dots, y_n$  be  $n$  solutions of the homogeneous linear  $n$ th-order differential equation on an interval  $I$ . Then the set of solutions is linearly dependent on  $I$  if and only if  $W(y_1, y_2, \dots, y_n) = 0$  for all  $x$  in the interval.

**Solved Problems:**

**Example-1** Show that the functions  $e^x, e^{2x}, e^{3x}$  are linearly independent.

**Solution** Let,  $f_1 = e^x, f_2 = e^{2x}, f_3 = e^{3x}$

$\therefore$  Wronskian of  $f_1, f_2, f_3$  is

$$W(f_1, f_2, f_3) = \begin{vmatrix} f_1 & f_2 & f_3 \\ f_1' & f_2' & f_3' \\ f_1'' & f_2'' & f_3'' \end{vmatrix}$$

$$\Rightarrow W(x) = \begin{vmatrix} e^x & e^{2x} & e^{3x} \\ e^x & 2e^{2x} & 3e^{3x} \\ e^x & 4e^{2x} & 9e^{3x} \end{vmatrix}$$

$$\Rightarrow W(x) = e^x \cdot e^{2x} \cdot e^{3x} \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix}$$

$$\Rightarrow W(x) = e^{6x} \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 5 \end{vmatrix} ; \begin{bmatrix} c_2' = c_1 - c_2 \\ c_2' = c_2 - c_3 \end{bmatrix}$$

$$\Rightarrow W(x) = e^{6x} \begin{vmatrix} 1 & 1 \\ 3 & 5 \end{vmatrix} = 2e^{6x} \neq 0, \text{ for all } x.$$

Since  $W(x) \neq 0$ , the functions  $e^x, e^{2x}, e^{3x}$  are linearly independent. **[Shown]**

**Example-2** Show that  $1, \sin^2 x, \cos^2 x$  are linearly independent.

**Solution** Let,  $f_1 = 1, f_2 = \sin^2 x, f_3 = \cos^2 x$

$\therefore$  Wronskian of  $f_1, f_2, f_3$  is

$$W(f_1, f_2, f_3) = \begin{vmatrix} f_1 & f_2 & f_3 \\ f_1' & f_2' & f_3' \\ f_1'' & f_2'' & f_3'' \end{vmatrix}$$

$$\Rightarrow W(x) = \begin{vmatrix} 1 & \sin^2 x & \cos^2 x \\ 0 & 2\sin x \cos x & -2\cos x \sin x \\ 0 & 2\cos^2 x - 2\sin^2 x & -2\cos^2 x + 2\sin^2 x \end{vmatrix}$$

$$\Rightarrow W(x) = 2\sin x \cos x (2\cos^2 x - 2\sin^2 x) \begin{vmatrix} 1 & \sin^2 x & \cos^2 x \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{vmatrix}$$

$$\Rightarrow W(x) = 4\sin x \cos x (\cos^2 x - \sin^2 x) \times 0 = 0, \text{ for all } x.$$

Since  $W(x) = 0$ , so  $1, \sin^2 x, \cos^2 x$  are linearly dependent. **[Shown]**

**Example-3** By determining constants  $c_1, c_2, c_3, c_4$  which are not all zero and are such that  $c_1 f_1 + c_2 f_2 + c_3 f_3 + c_4 f_4 = 0$  identically, show that the functions  $f_1(x) = x, f_2(x) = e^x, f_3(x) = xe^x, f_4(x) = (2-3x)e^x$  are linearly dependent.

**Solution** From the relation  $c_1 f_1 + c_2 f_2 + c_3 f_3 + c_4 f_4 = 0$ , we obtain

$$c_1 x + c_2 e^x + c_3 x e^x + c_4 (2-3x)e^x = 0$$

$$\Rightarrow c_1 x + (c_2 + 2c_4)e^x + (c_3 - 3c_4)xe^x = 0 \quad \dots (1)$$

Equating coefficients of like terms from both sides of (1), we get

$$c_1 = 0, c_2 + 2c_4 = 0, c_3 - 3c_4 = 0.$$

There are four variables having in three equations, so there is one free variable, which we let  $c_4$ . If we let  $c_4 = 1$  (you may let any real number, except  $c_4 = 0$ ), then we obtain  $c_3 = 3, c_2 = -2$ .

Thus one set of  $c$ 's is:  $c_1 = 0, c_2 = -2, c_3 = 3, c_4 = 1$  **(Ans.)**

**Example-4** Show that the functions  $x, x^2, x^3$  are linearly independent. If we consider these functions are three linearly independent solutions of a differential equation, find the differential equation.

**Solution**

Let,  $f_1 = x, f_2 = x^2, f_3 = x^3$

$\therefore$  Wronskian of  $f_1, f_2, f_3$  is

$$W(f_1, f_2, f_3) = \begin{vmatrix} f_1 & f_2 & f_3 \\ f_1' & f_2' & f_3' \\ f_1'' & f_2'' & f_3'' \end{vmatrix} \Rightarrow W(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$$

$$\Rightarrow W(x) = 6x^3 - 6x^3 + 2x^3$$

$$\therefore W(x) = 2x^3$$

Since  $W(x) \neq 0$ , for all  $x$ , so the functions  $x, x^2, x^3$  are linearly independent. **[Shown]**

**2nd Part :** Since,  $f_1, f_2, f_3$  are three linear independent solutions, we let its general solution is

$$\begin{aligned} y &= c_1 f_1 + c_2 f_2 + c_3 f_3 \\ \Rightarrow y &= c_1 x + c_2 x^2 + c_3 x^3 \end{aligned} \quad \dots (1)$$

where  $c_1, c_2, c_3$  are arbitrary constants.

Now differentiating (1) successively w. r. to  $x$ , we get

$$y_1 = c_1 + 2c_2 x + 3c_3 x^2 \quad \dots (2)$$

$$y_2 = 2c_2 + 6c_3 x \quad \dots (3)$$

$$\text{and } y_3 = 6c_3 \Rightarrow c_3 = \frac{1}{6} y_3 \quad \dots (4)$$

Putting value of  $c_3$  from (3) into (4), we get

$$\begin{aligned} y_2 &= 2c_2 + xy_3 \\ \therefore c_2 &= \frac{1}{2}(y_2 - xy_3) \end{aligned} \quad \dots (5)$$

Putting values of  $c_3$  and  $c_2$  in (5), we get

$$\begin{aligned} y_1 &= c_1 + x(y_2 - xy_3) + 3x^2 \cdot \frac{1}{6} y_3 \\ \therefore c_1 &= y_1 - xy_2 + \frac{1}{2} x^2 y_3 \end{aligned}$$

Now putting the values of  $c_1, c_2, c_3$  in (1), we obtain the desired differential equation

$$\begin{aligned} y &= x \left( y_1 - xy_2 + \frac{1}{2} x^2 y_3 \right) + \frac{1}{2} x^2 (y_2 - xy_3) + \frac{1}{6} x^3 y_3 \\ \Rightarrow 6y &= 6xy_1 - 6x^2 y_2 + 3x^3 y_3 + 3x^2 y_2 - 3x^3 y_3 + x^3 y_3 \\ \Rightarrow 6y &= 6xy_1 - 3x^2 y_2 + x^3 y_3 \\ \therefore x^3 y_3 - 3x^2 y_2 + 6xy_1 - 6y &= 0 \quad \text{(Ans.)} \end{aligned}$$

## Section-5.2

### Higher Order Linear Differential Equations

#### 5.1.1 General Linear Differential Equations

The differential equation of the form

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x) y = R(x) \quad \dots (1)$$

where the coefficients  $a_0(x), a_1(x), \dots, a_n(x)$  and function  $R(x)$  are to be independent of the variable  $y$ , is called general linear equation of order  $n$ .

If  $R(x) = 0$  is identically zero, differential equation (1) is said to be linear and homogeneous. If  $R(x)$  is not identically zero, differential equation (1) is said to be linear and non-homogeneous.

#### 5.1.2 Solution of Homogeneous Linear Differential Equations

Let  $y_1, y_2, \dots, y_n$  be any  $n$  solutions of the homogeneous linear differential equation

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x) y = 0 \quad \dots (2)$$

Then,  $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ , where  $c_1, c_2, \dots, c_n$  are  $n$  arbitrary constants, is also the solution of the homogeneous linear differential equation (2). Since, the solution contains  $n$  arbitrary constants; this solution is the general solution of the homogeneous linear differential equation (2).

#### 5.1.3 Solution of Non-Homogeneous Linear Differential Equations

Let  $y_1, y_2, \dots, y_n$  be any  $n$  solutions of the homogeneous linear differential equation

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x) y = 0 \quad \dots (1)$$

Then,  $y_c = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ , where  $c_1, c_2, \dots, c_n$  are  $n$  arbitrary constants, is called the **complementary solution or complementary function** of the linear differential equation (1).

Also let  $y_p$  be the particular solution of the non-homogeneous linear differential equation

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x) y = R(x) \quad \dots (1)$$

Then,  $y = y_c + y_p$  is the general solution of the non-homogeneous linear differential equation (1).

#### 5.2.4 General Linear Differential Equations with constant coefficients

The differential equation of the form

$$a_0(x) \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = R(x) \quad \dots (1)$$

where the coefficients  $a_0, a_1, \dots, a_n$  are all constants and function  $R(x)$  are to be independent of the variable  $y$ , is called general linear equation of order  $n$ .

If  $R(x) = 0$  is identically zero, the differential equation (1) is said to be linear and homogeneous. If  $R(x)$  is not identically zero, differential equation (1) is said to be linear and non-homogeneous.

### 5.2.5 General Solution of Homogeneous Linear Differential Equations with constant coefficients

Let  $y_1, y_2, \dots, y_n$  be any  $n$  solutions of the homogeneous linear differential equation

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = 0 \quad \dots (2)$$

Then,  $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ , where  $c_1, c_2, \dots, c_n$  are  $n$  arbitrary constants, is also the solution of the homogeneous linear differential equation (2). Since, the solution contains  $n$  arbitrary constants; this solution is the general solution of the homogeneous linear differential equation (2).

### 5.2.6 Differential Operator

Let  $D$  denote differentiation with respect to  $x$ ,  $D^2$  differentiation twice with respect to  $x$ , and so on; that is, for positive integral  $r$ ,

$$D^r y = \frac{d^r y}{dx^r}$$

The expression

$$A = a_0 D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n \quad \dots (1)$$

is called a differential operator of order  $n$ . If  $y$  is a function which is differentiable  $n$  times then, when the operator (1) is applied to the function  $y$ , yields the result

$$Ay = a_0 D^n y + a_1 D^{n-1} y + a_2 D^{n-2} y + \dots + a_{n-1} D y + a_n y \quad \dots (2)$$

$$\text{or, } Ay = a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y \quad \dots (3)$$

where the coefficients  $a_0, a_1, \dots, a_n$  in the operator  $A$  may be functions of  $x$ , constant.

### 5.2.7 Auxiliary Equation

Consider the differential equation

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = 0 \quad \dots (2)$$

$$\text{or, } (a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n) y = 0 \quad \dots (3)$$

where the coefficients  $a_0, a_1, \dots, a_n$  are all constants.

Let  $y = e^{mx}$  be a solution of the equation (2) or (3). Then we have,

$$Dy = \frac{dy}{dx} = me^{mx}, D^2 y = \frac{d^2 y}{dx^2} = m^2 e^{mx}, \dots, D^n y = \frac{d^n y}{dx^n} = m^n e^{mx},$$

Substituting these values in equation (1) or (2), we obtain the equation

$$(a_0 m^n + a_1 m^{n-1} + \dots + a_{n-1} m + a_n) e^{mx} = 0$$

Since  $e^{mx} \neq 0$ , hence  $e^{mx}$  will be a solution of (2) or (3) if  $m$  is a root of the algebraic equation

$$a_0 m^n + a_1 m^{n-1} + \dots + a_{n-1} m + a_n = 0 \quad \dots (4)$$

This equation is called the **Auxiliary equation** or the **characteristic equation** of the given differential equation.

### 5.2.8 Solution of Homogeneous Linear Differential Equations

To obtain general solution of a linear homogeneous differential equation with constant coefficients, we consider the three cases as below:

**Case I:** Distinct Real Roots i.e., when all the roots of auxiliary equation are real and different:

Suppose  $m_1, m_2, \dots, m_n$  are  $n$  distinct real roots of the auxiliary equation (4), then  $y = e^{m_1 x}, y = e^{m_2 x}, \dots, y = e^{m_n x}$  are all independent solutions of (2) or (3). Therefore, the general solution of the differential equation (2) or (3) is

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$$

where  $C_1, C_2, \dots, C_n$  are arbitrary constants.

**Example:** If roots of an auxiliary equation be  $m = -1, 1, 3, 4$  (i.e., four distinct real roots) then the solution will be  $y = C_1 e^{-x} + C_2 e^x + C_3 e^{3x} + C_4 e^{4x}$ .

**Case II:** Repeated Real Roots i.e., when all the roots of auxiliary equation are real and equal:

We have shown in case I, that is  $m_1, m_2, \dots, m_n$  are  $n$  distinct real roots, then the general solution is

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$$

But if  $m_1 = m_2$ , i.e., two roots are equal, then the above solution becomes

$$y = (C_1 + C_2) e^{m_1 x} + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}$$

It is clear from the above solution, it contains  $n-1$  arbitrary constants, since  $C_1 + C_2$  is equivalent to another only one arbitrary constant (say  $C_1 + C_2 = C$ ). So this is no longer a general solution.

To illustrate this we consider a second order linear differential equation having two equal roots of its auxiliary equation,

$$(D - m_1)^2 y = 0 \quad \dots (5)$$

We put  $(D - m_1)y = z$ , then equation (5) becomes

$$(D - m_1)z = 0$$

$$\Rightarrow \frac{dz}{dx} - m_1 z = 0, \text{ which is a separable equation, } \left[ \text{here, } D \equiv \frac{d}{dx} \right]$$

$$\Rightarrow \frac{dz}{z} - m_1 dx = 0$$

$$\Rightarrow \ln z - m_1 x = \ln C_2$$

$$\Rightarrow \ln z = \ln C_2 e^{m_1 x}$$

$$\Rightarrow z = C_2 e^{m_1 x}$$

$$\Rightarrow (D - m_1)y = C_2 e^{m_1 x} \quad [\because (D - m_1)y = z]$$

$$\Rightarrow \frac{dy}{dx} - m_1 y = C_2 e^{m_1 x}$$

which is a first order linear differential equation, so its integrating factor, I.F. =  $e^{\int (-m_1) dx} = e^{-m_1 x}$ .

Now, multiplying both sides of the above equation by  $e^{-m_1 x}$ , we obtain

$$e^{-m_1 x} \frac{dy}{dx} - m_1 e^{-m_1 x} \cdot y = C_2$$

$$\Rightarrow \frac{d}{dx} (y e^{-m_1 x}) = C_2$$

$$\Rightarrow \int d(ye^{-m_1x}) = \int C_2 dx$$

$$\Rightarrow ye^{-m_1x} = C_2x + C_1$$

$$\therefore y = (C_1 + C_2x)e^{m_1x}$$

Therefore, when two roots of the auxiliary equation (4) are equal and all other roots are distinct, i.e.,  $m_1 = m_2$  and  $m_3, m_4, \dots, m_n$  are  $n-2$  distinct real roots, then the general solution of the differential equation is  $y = (C_1 + C_2x)e^{m_1x} + C_3e^{m_3x} + \dots + C_ne^{m_nx}$

Hence, if all the roots are equal i.e.,  $m_1 = m_2 = \dots = m_n$  (say  $= m$ ), then the general solution of the differential equation is  $y = (C_1 + C_2x + C_3x^2 + \dots + C_nx^{n-1})e^{mx}$

where  $C_1, C_2, \dots, C_n$  are arbitrary constants.

### Examples:

- (i) If roots of an auxiliary equation be  $m = 2, 2, 2$  (i.e., three real and equal roots) then the solution will be  $y = (C_1 + C_2x + C_3x^2)e^{2x}$ .
- (ii) If roots of an auxiliary equation be  $m = -1, 2, 3, 3, 3$  (i.e., two real and distinct roots and three real and equal roots) then the solution will be  $y = C_1e^{-x} + C_2e^{2x} + (C_3 + C_4x + C_5x^2)e^{3x}$ .

### Case III: Conjugate complex roots:

Let  $a \pm ib$  be the conjugate complex roots of the auxiliary equation of a second order linear homogeneous differential equation. Since roots are distinct, we let  $m_1 = a + ib$  and  $m_2 = a - ib$ , then the general solution is

$$\begin{aligned} y &= C_1e^{(a+ib)x} + C_2e^{(a-ib)x} \\ \Rightarrow y &= C_1e^{ax} \cdot e^{ibx} + C_2e^{ax} \cdot e^{-ibx} \\ \Rightarrow y &= e^{ax} [C_1(\cos bx + i \sin bx) + C_2(\cos bx - i \sin bx)] \\ \Rightarrow y &= e^{ax} [(C_1 + C_2)\cos bx + (C_1 - C_2)i \sin bx] \\ \Rightarrow y &= e^{ax} [A \cos bx + B \sin bx] \end{aligned}$$

where  $A = C_1 + C_2$  and  $B = (C_1 - C_2)i$  are two new arbitrary constants.

i.e., if  $a \pm ib$  be the conjugate complex roots of the auxiliary equation of a second order linear homogeneous differential equation, then its general solution is  $y = e^{ax}(C_1 \cos bx + C_2 \sin bx)$ .

**Repeated conjugate complex roots:** If the auxiliary equation of a 4<sup>th</sup> order differential equation has two equal conjugate complex roots  $a \pm ib$ , i.e.,  $m_1 = a + ib, m_2 = a - ib, m_3 = a + ib$  and  $m_4 = a - ib$ , then its general solution will be

$$y = e^{ax} [(C_1 + C_2x)\cos bx + (C_3 + C_4x)\sin bx]$$

where  $C_1, C_2, C_3$ , and  $C_4$  are arbitrary constants.

### Examples:

- (i) If conjugate complex roots of an auxiliary equation be  $3 \pm 2i$ , i.e.,  $m_1 = 3 + 2i$  and  $m_2 = 3 - 2i$ , then the solution will be  $y = e^{3x}(C_1 \cos 2x + C_2 \sin 2x)$ .
- (ii) If conjugate complex roots of an auxiliary equation be  $\pm 2i$ , i.e.,  $m_1 = +2i$  and  $m_2 = -2i$  are pure imaginary, then the solution will be  $y = e^{0x}(C_1 \cos 2x + C_2 \sin 2x) = C_1 \cos 2x + C_2 \sin 2x$ .

(iii) If conjugate complex roots of an auxiliary equation be  $-1 \pm 4i$  repeated twice i.e.,  $m_1 = -1 + 4i$ ,  $m_2 = -1 - 4i$ ,  $m_3 = -1 + 4i$  and  $m_4 = -1 - 4i$ , then its general solution will be  $y = e^{-x}[(C_1 + C_2x)\cos 4x + (C_3 + C_4x)\sin 4x]$

### Solved Problems:

**Example-1** Solve the Differential Equation  $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$

**Solution** Given differential equation can be written as follows

$$(D^2 - 3D + 2)y = 0 \quad \dots (1)$$

Let  $y = e^{mx}$  be the trial solution of the differential equation (1), then get

$$Dy = me^{mx} \text{ and } D^2y = m^2e^{mx}$$

Putting these values in equation (1), we obtain

$$(m^2 - 3m + 2)e^{mx} = 0$$

Since  $e^{mx} \neq 0$ , the auxiliary equation of the homogeneous differential equation (1), is

$$m^2 - 3m + 2 = 0$$

$$\Rightarrow (m-1)(m-2) = 0$$

$$\therefore m = 1, 2$$

Therefore, the general solution of (1) is

$$y = C_1e^x + C_2e^{2x}. \text{ (Ans)}$$

**Example-2** Solve the Differential Equation  $\frac{d^3y}{dx^3} + 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} + 6y = 0$

**Solution** Given differential equation can be written as follows

$$(D^3 + 6D^2 + 11D + 6)y = 0 \quad \dots (1)$$

Let  $y = e^{mx}$  be the trial solution of the differential equation (1), then get

$$Dy = me^{mx}, D^2y = m^2e^{mx} \text{ and } D^3y = m^3e^{mx}$$

Putting these values in equation (1), we obtain

$$(m^3 + 6m^2 + 11m + 6)e^{mx} = 0$$

Since  $e^{mx} \neq 0$ , the auxiliary equation of the homogeneous differential equation (1), is

$$m^3 + 6m^2 + 11m + 6 = 0$$

$$\Rightarrow m^3 + m^2 + 5m^2 + 5m + 6m + 6 = 0$$

$$\Rightarrow (m+1)(m^2 + 5m + 6) = 0$$

$$\Rightarrow (m+1)(m^2 + 2m + 3m + 6) = 0$$

$$\Rightarrow (m+1)\{m(m+2) + 3(m+2)\} = 0$$

$$\Rightarrow (m+1)(m+2)(m+3) = 0$$

$$\therefore m = -1, -2, -3$$

Therefore, the general solution of (1) is

$$y = C_1e^{-x} + C_2e^{-2x} + C_3e^{-3x}. \text{ (Ans.)}$$



**Example-3** Solve the Differential Equation  $\frac{d^3y}{dx^3} - 2\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 8y = 0$

**Solution** Given differential equation can be written as follows

$$(D^3 - 2D^2 - 4D + 8)y = 0 \quad \dots (1)$$

Let  $y = e^{mx}$  be the trial solution of the differential equation (1), then get

$$Dy = me^{mx}, D^2y = m^2e^{mx} \text{ and } D^3y = m^3e^{mx}$$

Putting these values in equation (1), we obtain

$$(m^3 - 2m^2 - 4m + 8)e^{mx} = 0$$

Since  $e^{mx} \neq 0$ , the auxiliary equation of the homogeneous differential equation (1), is

$$m^3 - 2m^2 - 4m + 8 = 0$$

$$\Rightarrow m^3 + 2m^2 - 4m^2 - 8m + 4m + 8 = 0$$

$$\Rightarrow m^2(m + 2) - 4m(m + 2) + 4(m + 2) = 0$$

$$\Rightarrow (m + 2)(m^2 - 4m + 4) = 0$$

$$\Rightarrow (m + 2)(m - 2)^2 = 0$$

$$\therefore m = -2, 2, 2$$

Therefore, the general solution of (1) is

$$y = C_1e^{-2x} + (C_2 + C_3x)e^{2x}. \text{ (Ans)}$$

**Example-4** Solve the Differential Equation  $(D^4 + 5D^2 + 6)y = 0$

**Solution** Given differential equation

$$(D^4 + 5D^2 + 6)y = 0 \quad \dots (1)$$

Let  $y = e^{mx}$  be the trial solution of the differential equation (1), then the auxiliary equation of (1), is

$$m^4 + 5m^2 + 6 = 0$$

$$\Rightarrow m^4 + 3m^2 + 2m^2 + 6 = 0$$

$$\Rightarrow m^2(m^2 + 3) + 2(m^2 + 3) = 0$$

$$\Rightarrow (m^2 + 3)(m^2 + 2) = 0$$

$$\therefore m = \pm i\sqrt{2}, \pm i\sqrt{3}$$

Therefore, the general solution or complete solution of (1) is

$$y = C_1 \cos \sqrt{2}x + C_2 \sin \sqrt{2}x + C_3 \cos \sqrt{3}x + C_4 \sin \sqrt{3}x. \text{ (Ans.)}$$

**Example-5** Solve the Differential Equation  $(D^4 - D^3 - D + 1)y = 0$

**Solution** Given differential equation

$$(D^4 - D^3 - D + 1)y = 0 \quad \dots (1)$$

Let  $y = e^{mx}$  be the trial solution of the differential equation (1), then the auxiliary equation of (1), is

$$m^4 - m^3 - m + 1 = 0$$

$$\Rightarrow (m-1)(m^3-1)=0$$

$$\Rightarrow (m-1)(m-1)(m^2+m+1)=0$$

$$\therefore m=1, 1 \text{ and } m^2+m+1=0$$

$$\Rightarrow m = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$\Rightarrow m = \frac{-1 \pm \sqrt{-3}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$$

Therefore, the complete solution of (1) is

$$y = (C_1 + C_2 x)e^x + e^{\frac{-x}{2}} \left( C_3 \cos \frac{\sqrt{3}}{2} x + C_4 \sin \frac{\sqrt{3}}{2} x \right) \text{ (Ans.)}$$

**Example-6** Solve the Differential Equation

$$(D^2 - 6D + 25)y = 0, y(0) = -3, y'(0) = -1$$

**Solution** Given differential equation

$$(D^2 - 6D + 25)y = 0 \quad \dots (1)$$

and initial conditions

$$y(0) = -3 \quad \dots (2)$$

$$y'(0) = -1 \quad \dots (3)$$

Let  $y = e^{mx}$  be the trial solution of the differential equation (1), then the auxiliary equation of (1), is

$$m^2 - 6m + 25 = 0$$

$$\Rightarrow m = \frac{-(-6) \pm \sqrt{(-6)^2 - 4 \cdot 1 \cdot 25}}{2 \cdot 1}$$

$$\Rightarrow m = \frac{6 \pm \sqrt{-64}}{2} = \frac{6 \pm 8i}{2} = 3 \pm 4i$$

Therefore, the complete solution of (1) is

$$y(x) = e^{3x}(C_1 \cos 4x + C_2 \sin 4x) \quad \dots (4)$$

We also have  $y'(x) = 3e^{3x}(C_1 \cos 4x + C_2 \sin 4x) + e^{3x}(-4C_1 \sin 4x + 4C_2 \cos 4x) \dots (5)$

Using initial condition  $y(0) = -3$ , i.e., putting  $x = 0, y = -3$  in equation (4), we get

$$y(0) = C_1 \cdot 1 \Rightarrow C_1 = -3$$

Again using initial condition  $y'(0) = -1$ , i.e., putting  $x = 0, y' = -1$  in equation (5), we get

$$y'(0) = 3(C_1 \cdot 1 + C_2 \cdot 0) + (-4C_1 \cdot 0 + 4C_2 \cdot 1)$$

$$\Rightarrow -1 = 3C_1 + 4C_2$$

$$\Rightarrow -1 = -9 + 4C_2$$

$$\therefore C_2 = 2$$

Putting values of  $C_1$  and  $C_2$  in equation (4), we obtain the required solution

$$y(x) = e^{3x}(-3 \cos 4x + 2 \sin 4x) \text{ (Ans.)}$$